Clinical Simulation in Nursing 105 (2025) 101782

Contents lists available at ScienceDirect

Clinical Simulation in Nursing

journal homepage: www.elsevier.com/locate/ecns

Research Article

Artificial intelligence (AI)-facilitated debriefing: A pilot study

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ARTICLE INFO

KEYWORDS:
Artificial intelligence
Debrief
Nursing
Reflective practice
Simulation

ABSTRACT

Background: Debriefing is a cornerstone of simulation-based education (SBE), enabling reflective practice to enhance learning outcomes. Artificial intelligence (AI)-facilitated debriefing is an emerging innovation with limited research in nursing education.
Purpose: This study explored the relationship between time spent in an AI-facilitated debrief, the number of reflective dimensions met, and student performance in a virtual nursing simulation.
Methods: A mixed-methods approach integrated quantitative data (simulation scores, time in debrief, number of dimensions met) and qualitative insights from AI algorithm. Participants ($N = 52$) completed a screen-based simulation and an AI-facilitated debrief guided by the EMPOWER® Debriefing Framework. Descriptive statistics, Pearson's correlations, and t-tests were conducted; qualitative data underwent thematic analysis.
Results: No significant correlation was found between time in debrief and grades ($r = 0.07$, $p = .46$). Students meeting more dimensions (3-4) spent significantly less time in debrief ($M = 9.39$ min) than those meeting fewer dimensions (1–2) ($M = 12.62$ min), $t(50)=5.43$, $p < .001$.
Conclusions: AI-facilitated debriefing shows potential for scalable reflective practice but may not replace the depth of human-facilitated sessions. Integration into nursing education requires further validation of reflective outcomes.

Background and significance

In education, emotions play a pivotal role in the learning process, directly influencing students' engagement and knowledge retention (Thomas & Allen, 2021). Madsgaard et al. (2022) found that simulation-based education (SBE) evokes a wide range of co-existing and shifting emotions among nursing students, with unpleasant emotions often decreasing over time. Understanding these emotional dynamics offers valuable insights for nurse educators. Similarly, Salo et al. (2025) reported that while SBE provides realistic clinical experiences that enhance learning and self-confidence, the emotional impact on learners remains underexplored. Their qualitative study, which analyzed diary entries and online chats from 56 Finnish nursing students across 32 simulation scenarios, revealed that although simulations elicited both positive and negative emotions, unpleasant feelings such as anxiety and stress predominated. This highlights the importance of acknowledging and managing emotions to improve the effectiveness and satisfaction of SBE.

Positive emotions like excitement can foster motivation, while negative emotions like anxiety may impair concentration and learning efficacy. Simulation-based education (SBE) creates an immersive environment, with debriefing sessions serving as critical reflective tools to assess both cognitive and emotional responses (Dreifuerst, 2009). Understanding students' sentiments during these sessions can help enhance debriefing strategies and inform curriculum development, leading to more tailored and emotionally attuned educational experiences. Moreover, the integration of voice-based AI technologies, particularly Large Language Models (a type of neural network), into debriefing processes offers promising support for cognitive emotion analysis. Given the close connection between emotions, thoughts, behaviors, and decision-making, providing real-time feedback on student reflections through audio recordings may help bridge gaps for asynchronous learners.

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https://doi.org/10.1016/j.ecns.2025.101782

Debriefing in simulation-based learning

Debriefing is widely regarded as one of the most essential components of simulation-based education (SBE) (International Nursing Association for Clinical Simulation and Learning [INACSL], 2021). It serves as the "aha" moment for both facilitators and learners, where insights are gained, and connections are made. The INACSL Healthcare Simulation Standards of Best Practice® (2021) define debriefing as a reflective process immediately following the SBE that is led by a trained facilitator using an evidence-based debriefing model. It is a deliberate, structured conversation where learners engage in metacognitive reflection on their performance (Dreifuerst, 2009). Best practice guidelines recommend that debriefing should occur immediately after the SBE session and follow a structured format, as structured debriefs have been shown to positively impact learning outcomes for nursing students (Lee et al., 2020). Research suggests that using a rubric during debriefing not only enhances critical thinking but also ensures a consistent experience, regardless of the facilitator's debriefing skills (Wong et al., 2020).

In an online asynchronous environment opportunity for debriefing can be limited due to geographical and time constraints, as well as the availability of both learners and faculty. To date, there are several ways to meet debriefing standards in hybrid or online settings (Verkuyl et al., 2020). These include self-directed debriefs followed by synchronous sessions with a facilitator or scheduling synchronous debriefs at mutually convenient times. While immediate self-debriefs can be valuable, as acknowledged by Verkuyl et al. (2020), they may lack the broader insights that faculty can gain from reviewing aggregate student performance.

Artificial intelligence (AI) and debriefing

The use of AI-facilitated debriefing in nursing education is still a relatively new and novel approach, and as a result, the literature on this topic remains limited. However, industries outside of healthcare have seen greater adoption of this technology for purposes such as interviews and critical conversations (Hunkenschroer & Luetge, 2022). Within nursing education, AI applications are steadily increasing, uses include auto-grading, case development, and item writing (Gonzalez, 2024).

Brutschi et al. (2024) explored the feasibility of using audio-based speaker diarization technology to automatically measure and visualize interaction patterns during debriefings in simulation-based medical education. Audio recordings from debriefings with medical students were processed using machine learning algorithms to generate visual interaction maps (sociograms) with high accuracy (97.78%). The findings demonstrated that this technology can objectively assess debriefing dynamics, providing valuable insights for improving faculty development and enhancing the effectiveness of simulation-based education.

Benfatah et al. (2024) conducted a randomized controlled trial ($N = 40$) to examine the effects of AI-facilitated debriefing on nursing students. Simulation sessions were recorded and analyzed using AI technology, which then delivered personalized feedback. The study found that the group receiving AI- facilitated debriefing showed significant improvements in participation and critical reflection ($p < .05$).

AI thematic analysis

The use of large language models (LLMs) such as ChatGPT for qualitative data analysis is a relatively new and evolving area of inquiry. As researchers explore the affordances and limitations of AI-assisted approaches, questions around methodological rigor, transparency, and ethical use remain central. Zhang et al. (2024) examined how ChatGPT (GPT–3.5) (OpenAI, 2023) can assist with thematic analysis of qualitative data. Through interviews and prompt-engineering workshops, the study developed an eight-element prompt framework encompassing context-setting, methodology, data input/output format, role assignment, transparency, prioritization, encouragement, and iterative refinement. The findings suggest that when prompts are carefully designed, ChatGPT can produce outputs with improved transparency, reproducibility, and coherence. Importantly, participants' attitudes shifted from initial skepticism to conditional trust as AI outputs became more interpretable. However, the study also highlights ethical concerns, including the risks of embedded bias, the need for human oversight, and the potential for researchers to overtrust AI-generated interpretations. Christou (2024) similarly emphasizes that while AI can extend the scope, speed, and accessibility of thematic analysis, its use must be paired with critical human interpretation, transparency, and methodological fidelity.

Voice sentiment analysis in education

Voice sentiment analysis, also known as speech sentiment analysis, utilizes artificial intelligence (AI) and machine learning to analyze spoken language and determine the emotional tone or sentiment expressed by the speaker. Unlike traditional text-based sentiment analysis, this approach evaluates both the content of the speech (what is said) and the audio features (how it is said) to interpret emotions (Anand & Patra, 2022). In digital or remote learning environments, voice sentiment analysis can provide educators with insights into student engagement, motivation, or confusion based on their verbal interactions. Elbourhamy (2024) studied vision-impaired students who recorded themselves speaking, self-assessing, and reflecting on learning content. The AI model effectively identified at-risk students by predicting emotional indicators from these recordings. Voice sentiment analysis has significant potential to enhance educational processes by providing insights into learning outcomes and predicting academic performance (Elbourhamy, 2024). By understanding the emotional context behind students' verbal participation, educators can make more informed decisions about instructional strategies and support interventions.

Relativ.ai Inc. (2023) has developed a cutting-edge conversational intelligence solution that leverages customized large language models and machine learning algorithms to extract actionable insights from spoken dialogue (Relativ.ai Inc., 2023). By applying principles of psychology to inform their approach, they have designed a customized framework for analyzing linguistic features to diagnose conversation dynamics. Utilizing these features, their proprietary AI-powered tool generates feedback reports that provide granular insights into conversations, enabling users to identify areas of improvement and optimize their communication strategies. By integrating domain-specific knowledge with machine learning-driven analysis, Relativ.ai, Inc. (2023) has created a unique platform for conversational intelligence that fosters more effective dialogue, improved relationships, and enhanced decision-making. Relativ.ai, Inc. (2023) deploys customized AI models in educational institutions for career readiness preparation, in hospitality for customer service, in recruitment to provide job advice, and in corporate settings to help with upskilling workforces in the areas of leadership, client success, and team collaboration. With the right training (input) data, it excels at identifying traits such as cognitive skills, social skills, leadership characteristics, and emotional intelligence, alongside job-specific technical abilities, to provide actionable insights into performance.

Theoretical framework

This pilot study is grounded in Schön's Reflective Practice Theory (Schön, 1983) which emphasizes the role of reflection in professional development. In the context of nursing education, reflective practice is a critical component of learning, particularly in simulation-based environments where learners engage in reflection-in-action and reflection-on-action. This study analyzes AI-facilitated debrief audio transcripts from learners following SBE, focusing on key dimensions such as clarity, authority, influence, critical thinking, and confidence. These dimensions represent how effectively students reflect on their experiences, articulate their thought processes, and demonstrate critical thinking during debriefing sessions. By correlating these reflective practices with academic performance, this study seeks to understand how deeper reflection and more effective communication during debriefs influence learner outcomes.

Purpose

This study explored the integration of Artificial Intelligence (AI) through the real-time analysis of spoken reflections from nursing students during debriefing. Using AI to analyze students' sentiments, emotions, and concerns, the study sought to enrich the simulation-based learning process, providing educators and facilitators with immediate insights into their students' emotional and cognitive landscapes. The goal was to merge qualitative insights with quantitative performance metrics, offering a dual approach that enhances the overall understanding of learning outcomes.

Hypothesis: AI-facilitated debrief will correlate time in debrief and nursing students' simulation scores.

Methodology

For this pilot study, we collaborated with Relativ.ai Inc. (2023) to explore AI-driven sentiment and emotion analysis to process students' spoken reflections during AI-facilitated debriefings. This study marks the first application of Relativ.ai Inc. (2023) within a structured debriefing framework.

We utilized our proprietary EMPOWER® Debrief Framework (Sentinel U, n.d.). EMPOWER® is an acronym that outlines a structured debriefing process to guide learners through a reflective exercise. Each component of the acronym addresses a critical element of debriefing. For example, E stands for *Explore*, prompting learners to share their initial impressions of the learning experience, while M stands for *Misconceptions*, encouraging reflection on pre-existing knowledge, assumptions, values, and beliefs. P stands for *performance standards and practice guidelines*. O stands for *Outcomes and Objectives*. The W stands for *Wins.* The E stands for *Evaluate,* and lastly R stands for *Reflect and Review*. Structured debriefing is most effective for improving learning outcomes, critical thinking, and clinical judgment (Lee et al., 2020). In addition, Self-led structured debriefs are a viable option when facilitator debriefs are unfeasible (MacKenna et al., 2021)

Relativ.ai Inc. (2023) employs two AI models in tandem during every session. The first model functions as an AI agent that guides learners through a structured debriefing, focusing on the individual's experience and performance. In collaboration with the data scientists at Relativ.ai Inc (2023), we developed scripts and prompts aligned with each component of the EMPOWER® acronym. This process encompasses practice reflection, learning optimization, and evaluation of successes and areas for improvement.

The second AI model from Relativ.ai Inc (2023) analyzes the conversational data generated during the debriefing, facilitated by the first AI model. In addition to content analysis, the model is capable of assessing the conversation across several customizable dimensions, including confidence, fluency, authority, critical thinking, and clarity, providing insights to inform and enhance future learning experiences. For the purposes of this study, we used the following definitions:

- Clarity- The ability to communicate clearly when communicating symptoms, explaining treatments or educating patients and families.
- Fluency- Smooth speech, appropriate pauses, adapting to audience, using connectives.
- Critical Thinking- Refers to the generation of new ideas or concepts to form new and meaningful associations.
- Authority- The ability to be compelling and persuasive.
- Confidence- Clear speech, clarity, appropriate volume, and tone.

Research design

This study employed a mixed-methods approach, integrating quantitative data (academic performance, defined as simulation scores) and qualitative data (AI-generated thematic reflections) to comprehensively evaluate the impact of AI-facilitated debriefing on student learning outcomes. This design allowed for the measurement of key variables while also providing rich, contextual insights into underlying learner patterns and reflective experiences.

Variables: (a) Time: The duration each student spent engaging with the AI agent was recorded. (b) Score: Performance scores from the virtual simulation were collected to determine the baseline competency of learners before the debriefing. (c) Dimensions: Using feedback generated by Relativ.ai Inc (2023), the study measured how many of the predefined dimensions (confidence, fluency, authority, critical thinking, and clarity) were effectively demonstrated by each learner during the debriefing.

Data analysis

The statistical methods used in this study were selected to evaluate relationships and differences among key variables. A Pearson correlation analysis was performed to assess the strength and direction of the relationship between continuous variables, such as time spent in the AI-facilitated debrief and student grades. Additionally, independent samples t-tests were employed to compare the means of continuous variables, such as grades and time spent in the AI-facilitated debrief, between independent groups of students categorized by the number of dimensions met during the debrief. These statistical tests were appropriate for the data and research questions, enabling a comprehensive examination of relationships and group differences.

Participants

Participants were nursing students ($N = 52$) from a large southwest university in the United States in their medical-surgical course of a BSN program. The AI-facilitated debrief intervention was a requirement after the completion of a virtual simulation. The simulation focused on prioritization of care. Students in the cohort use simulation regularly and debrief as part of best practice (INACSL, 2021).

Data collection

This study was conducted with institutional review board (IRB) approval to ensure ethical standards for research involving human participants. Participants engaged in the Prioritization of Care™ Advanced Medical-Surgical simulation from Sentinel U®, a screen-based simulation designed to teach learners how to prioritize care settings, patients, and interventions. Upon completing the simulation, participants received a comprehensive activity report containing their scores, responses, and feedback.

To initiate the AI-facilitated debrief, a hyperlink was embedded in the activity report. This link directed participants to Relativ.ai Inc. (2023) where they engaged with the AI agent that facilitated the debrief following the EMPOWER® Debrief Framework. As part of the study participants were provided with detailed instructions and a how-to sheet to ensure they understood how to interact with the AI Agent.

The debrief process was structured around the EMPOWER® Debriefing Framework, with each letter representing a core component of reflection and learning. The AI agent posed targeted questions, prompting participants to record verbal responses for each segment of the debrief. The debrief concluded with participants receiving an immediate report from Relativ.ai Inc. (2023) summarizing their performance. All personally identifiable information (PII) was de-identified. Relativ.ai Inc. (2023) did not have access to student names or emails, and only the principal investigator merged names and emails after the study. Data security was maintained through strict access control and encrypted storage.

Results

Analysis 1: Time spent in AI-facilitated debrief and simulation score

The analysis yielded a correlation coefficient of $r = 0.070$, indicating a very weak positive relationship between the time students spent in the AI-facilitated debrief and their grades. The p-value for this correlation was 0.463 ($p > .05$), suggesting that the relationship was not statistically significant. This means that there was no clear evidence that spending more time in the AI-facilitated debrief was associated with higher grades.

Analysis 2: Relationship between qualitative debriefing performance and simulation score

Students were grouped into two categories based on the number of qualitative performance dimensions met during the debrief. The dimensions were confidence, fluency, authority, critical thinking, and clarity. For this pilot study, *fluency* was excluded due to a lack of variability across the sample. This resulted in two groups: This resulted in two groups:

- Fewer dimensions (1-2).
- More dimensions (3-4).

We then conducted independent t-tests to compare simulation score, and time spent in simulation between these two groups. The comparison of grades between the two groups revealed no statistically significant difference. Students who met more dimensions had slightly higher average grades (94.50) compared to those who met fewer dimensions (88.74), but this difference was not significant ($p = .203$). This result suggests that meeting more dimensions does not necessarily result in better grades**,** indicating that high performance in the debrief may not directly correlate with higher simulation scores.

Analysis 3: Relationship between time spent in AI-facilitated debrief and number of dimensions met

The analysis of time spent in the AI-facilitated debrief showed a statistically significant difference between the groups: Students who met fewer dimensions spent significantly more time in the debrief (12.62 minutes on average) compared to those who met more dimensions (9.39 minutes on average) as shown in Table. The highly significant p-value ($p = 2.73\text{e-}06$) indicates a strong relationship, suggesting that students who met more dimensions tended to be more efficient in their engagement with the AI-facilitated debrief, completing it in less time. This finding implies that students who can demonstrate higher competency across multiple dimensions are also more focused and efficient, requiring less time to articulate their thoughts and reflections.

Table
Dimensions Met v. Time in Simulation and Grade

Dimensions: Confidence, Authority, Critical thinking & Clarity	Mean Grade (%)	Mean Time (min)
1-2 Dimensions Met	88.74	12.62
3-4 Dimensions Met	94.50	9.39

Note. Dimensions were derived from debriefing transcripts and evaluated Relativ.ai Inc. algorithm.

Qualitative findings

The AI-generated thematic analysis provided aggregate insights into learners' reflections following the AI-facilitated debrief. These synthesized responses, generated by Relativ.ai Inc (2023) natural language processing model, reflect common patterns in sentiment and content across participants. One representative excerpt summarized the collective responses as follows:

*The survey responses indicated a generally positive sentiment among participants regarding their clinical decision-making and prioritization skills. While many felt confident in their application of frameworks like ABCs, challenges in distinguishing urgent from non-urgent interventions were common. Misconceptions about prioritization and the urgency of conditions were noted, highlighting a need for clearer guidelines. Participants expressed a desire for further practice and training, particularly in understanding patient conditions and effective communication (*Relativ.ai Inc. 2023*).*

Another thematically aligned excerpt reinforced this finding:

*The survey responses reveal a strong confidence among participants in their clinical decision-making, yet highlight significant challenges in prioritization, particularly in complex scenarios. Many respondents acknowledged misconceptions regarding the urgency of medical conditions, emphasizing the need for improved understanding and application of decision-making frameworks. Participants expressed a desire for further training and hands-on simulations to enhance their skills (*Relativ.ai Inc. 2023*).*

These AI-synthesized excerpts suggest a consistent tension between learners perceived confidence and their demonstrated difficulty in prioritizing care, especially under complex conditions. They also underscore a recurring theme: the learners' recognition of the value of additional practice, guided simulation, and structured decision-making frameworks to support clinical judgment development.

Discussion

The original hypothesis, AI-facilitated debrief will correlate with time in debrief and nursing students' simulation scores was rejected. There was no relationship between student performance and time spent in debrief or in simulation. However, the findings from this study offer nuanced insights into the relationship between time spent in AI-facilitated debriefs, student performance scores and dimensions of critical thinking and articulation.

Performance without dimensional depth

Students who achieved high scores but did not meet many dimensions may demonstrate superficial knowledge. Although they performed well on assessments, their inability to articulate and synthesize information during debriefs suggests a lack of deeper understanding and critical engagement. This highlights a key inference; high scores alone do not equate to deep learning. Critical thinking and the ability to connect and articulate ideas are vital for true comprehension, which some high-scoring students lacked despite their performance. These findings are supported in part by the works of Benner's from Novice to Expert (1984) which puts forth the idea that learners can do well superficially without achieving deeper understanding. Tanner (2006) also describes surface level learning or pattern recognition without deep reasoning limits clinical decision making.

Debriefing dimensions do not predict simulation score

The analysis showed no statistically significant difference in scores between students who met fewer (1-2) versus more (3-4) dimensions. Although those who met more dimensions had slightly higher average grades, the difference was not conclusive, suggesting that meeting multiple dimensions does not guarantee better grades. This aligns with scholarship suggesting that traditional grading often fails to capture metacognitive processes or deep learning (Gikandi et al., 2011) and may explain why students can perform well academically without demonstrating comprehensive understanding during debriefing.

Cognitive efficiency and reflective depth in debriefing

Students who met more dimensions (3-4) not only spent less time in the AI-facilitated debrief but also demonstrated greater cognitive clarity and the ability to express reflections concisely. This efficiency suggests a higher degree of focus, streamlined thinking, and critical engagement during the debriefing process, consistent with findings in cognitive science and nursing expertise literature that link concise explanation with deeper internalized understanding (Benner, 1984; Chi et al., 1989). Early work by Chi et al. (1989) posits that poor students do not generate sufficient self-explanations, monitor their learning inaccurately and rely heavily on examples. This pattern may also be interpreted through the lens of cognitive load theory (Sweller, 1988), which posits that learners with more developed schemas can process and retrieve information more efficiently, allowing them to engage more meaningfully in reflection with less cognitive effort.

Recommendations for nursing education

To enhance nursing education, structured debriefing sessions should be prioritized as both reflective and assessment tools. Scripted debriefs, such as the EMPOWER™ Debrief Framework (Sentinel U, n.d.), are particularly effective when combined with AI-driven technologies. These tools not only guide reflection but also provide valuable insights into students' critical thinking, communication, and articulation skills. Importantly, AI-facilitated debriefing offers a scalable and flexible solution for asynchronous learning environments, where traditional instructor-led debriefs may not always be feasible.

These recommendations are not unique to nursing but healthcare education in general. Educators could leverage AI-facilitated debriefing to evaluate multiple dimensions of student performance, including clarity, authority, and critical thinking. By providing immediate feedback and actionable insights, AI debriefing systems can help bridge the gap in asynchronous settings, ensuring that all learners receive individualized, structured opportunities for reflection and skill development. These exercises may also serve as an indicator of deeper mastery, as students demonstrating higher competency often engage more effectively with the AI-facilitated debriefing process.

Furthermore, educators could use these tools to identify and support students who may excel in traditional assessments but require additional guidance in critical thinking and articulation. Feedback should be structured to encourage deeper engagement with the material and to help students move beyond surface-level knowledge. By integrating AI-facilitated debriefing systems into nursing curricula, educators can foster a more holistic and accessible approach to learning, ensuring students are prepared for the complexities of real-world clinical environments, regardless of whether they learn synchronously or asynchronously.

Limitations

This study has several limitations that should be acknowledged. First, the findings are not generalized beyond the Southwest BSN school of nursing that participated. The sample was drawn from a specific educational setting, and results may vary in different contexts, institutions, or regions. Additionally, the study was underpowered; the sample size was smaller than optimal for detecting more subtle effects, which may have limited the ability to draw robust conclusions.

Another limitation is that the dimensions used to assess student performance during the AI-facilitated debrief were not developed by the principal investigator (PI); they were predefined by Relativ.ai Inc. (2023). While these dimensions provided a useful framework for this pilot study, they may not fully capture the unique competencies needed for nursing learners. Relativ.ai Inc (2023) can customize measures and uses a proprietary process to train their customized AI models to provide feedback during these conversations. Future research will aim to refine and better tailor these dimensions to the specific skills and learning outcomes relevant to nursing education. Further exploration into the validity of these dimensions would also enhance the robustness of the assessment.

Despite these limitations, this pilot study represents a valuable starting point for integrating AI-facilitated debriefing in simulation. It provides a foundation for future work to improve and adapt the assessment criteria, ensuring that they are more reflective of the competencies required for nursing professionals.

Conclusion

This study aimed to explore the relationship between scores on an SBE and the experience of learners with AI facilitated debriefs. There was no statistical relationship observed. However, this study did find that high scores do not always equate to deep learning or critical thinking. Students who met more dimensions and spent less time in debriefs demonstrated greater efficiency and focus, suggesting a deeper mastery of the material. Conversely, those who scored well but met fewer dimensions may have a superficial understanding, indicating a need for development in critical reflection and communication.

The use of AI generated themes is a nascent area of research. It is gaining traction among qualitative researchers as a reputable and reliable technology. AI-facilitated debriefing should not replace human interaction, but rather serve to augment structure, efficiency, and learner engagement. The optimal approach preserves psychological safety and reflective depth by combining AI insights with human-led dialogue.

These findings emphasize the importance of assessing beyond scores; critical thinking, efficiency, and the ability to articulate are crucial for professional success. Educators should focus on how learners engage with multiple dimensions to better develop these essential skills. This research encourages the integration of reflective practices and communication training in curricula, ensuring students are not only knowledgeable but also adept at applying their knowledge effectively in clinical settings.

For curriculum designers, simulationists, and educators, these insights offer a more comprehensive view of student readiness. Future research includes further exploration of the relationship between time efficiency, dimensions of performance, and long-term competency in clinical practice.

Declaration of competing interest

Laura Gonzalez is the VP for Innovation @ SentinelU, Arjun Nagendran is the Co-Founder at Relativ.

CRediT authorship contribution statement

Laura Gonzalez: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Arjun Nagendran:** Writing – review & editing, Software, Resources, Investigation.

Declaration of generative AI and AI- facilitated technologies in the writing process

During the preparation of this work the author(s) used ChatGPT to improve legibility. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication

Acknowledgments

I would like to acknowledge Arjun Nagendran-Relativ.ai Inc. (2023) collaborating with Sentinel U® on this unique study. I also want to acknowledge the nursing students and faculty who participated in this study, as well as the Innovation and Engineering teams who supported the integration of AI technology.

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